Math 520, F15 **Name:** Answer

Quiz 1 **SSN: xxx-xx-**

Instructions: **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (4 points) Verify that $y(t) = 3t^{1/2}$ is a solution of the differential equation

$$2t^2y'' + 3ty' - y = 0, \quad t > 0.$$

$$y(t) = 3t^{1/2}, \quad y'(t) = \frac{3}{2}t^{-1/2}, \quad y''(t) = -\frac{3}{4}t^{-3/2}$$

$$\Rightarrow 2t^2y'' + 3ty' - y = (2t^2)\cdot\left(-\frac{3}{4}t^{-3/2}\right) + 3t\cdot\left(\frac{3}{2}t^{-\frac{1}{2}}\right) - 3t^{1/2}$$

$$= -\frac{3}{2}t^{1/2} + \frac{9}{2}t^{1/2} - 3t^{1/2}$$

$$= 0$$

Problem 2. (6 points) Solve the following initial value problem:

$$t\frac{dy}{dt} + 2y = e^t, \qquad y(1) = 0, \qquad t > 0$$

standard form: $\dfrac{dy}{dt} + \left(\dfrac{2}{t}\right)y = \dfrac{e^t}{t}$

$$P(t) = \frac{2}{t}, \quad g(t) = \frac{e^t}{t}$$

Integrating factor: $\mu(t) = e^{\int \frac{2}{t}dt} = e^{2\ln t} = t^2$

Thus we have

$$\frac{d}{dt}[t^2y] = \frac{e^t}{t}\cdot t^2$$

$$t^2y = \int te^t\,dt = te^t - \int e^t dt = (t-1)e^t + C$$

$$\Rightarrow y(t) = \frac{t-1}{t^2}e^t + \frac{C}{t^2}$$

By the initial condition $y(1)=0$. we get

$$0 = 0 + C \Rightarrow C = 0$$

Thus the solution of the IVP is $y(t) = \left(\dfrac{t-1}{t^2}\right)e^t$